

BTECH II YEAR I SEMESTER

DEPARTMENT

Name of the Course

Course Code

Name of the Faculty

: Basic Science and Humanities

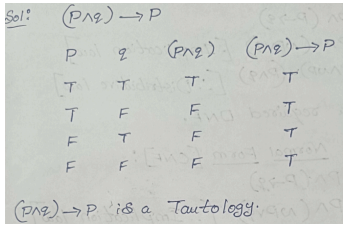
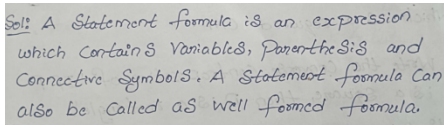
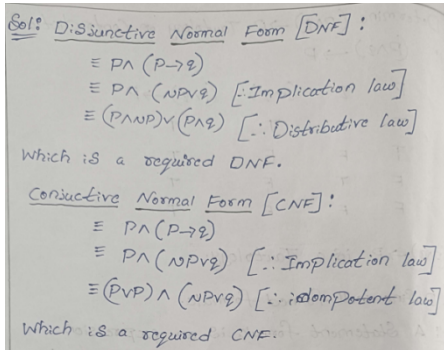
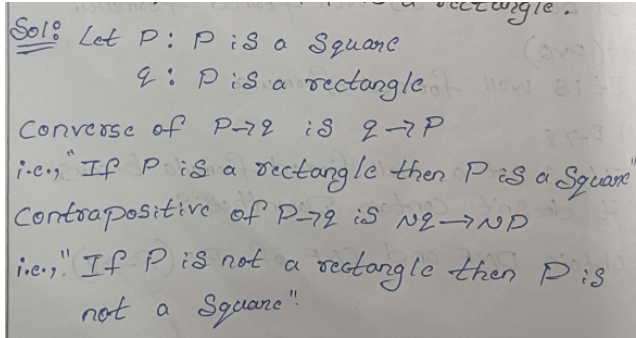
: Discrete Mathematics and Graph theory

: AR23BS108

: P. Rama Krishna

UNIT 1

Mathematical Logic: Propositional Calculus: Statements and Notations, Connectives, Well Formed Formulas, Truth Tables, Tautologies, and Equivalence of Formulas, Duality Law, Tautological Implications, and Normal Forms, Theory of Inference for Statement Calculus, Consistency of Premises and Indirect Method of Proof. Predicate Calculus: Predicative Logic, Statement Functions, Variables and Quantifiers, Free and Bound Variables, Inference Theory for Predicate Calculus.

S.No	QUESTIONS	YouTube Links
1	Determine $(P \wedge q) \rightarrow p$ Tautology or Contradiction? 	https://youtu.be/Ag187ZUMou0?si=_BBIcjFXkhoVSzqa
2	Define well formed formula 	https://youtu.be/e9BLD0ipL2M?si=_b7adVq14LJzh8sd
3	Obtain DNF and CNF of $p \wedge (p \rightarrow q)$. 	https://youtu.be/eYR6qcdeJRI?si=-k3cTZnhvne0HTn4
4	Write the converse, contrapositive of "If P is a square, then P is a rectangle". 	https://youtu.be/9mVRLkKoi7Q?si=bbaEfyBqCmzEgN4d

5	Without constructing the Truth Table, prove that $\sim (p \wedge q) \rightarrow (\sim p \vee (\sim p \vee q))$ and $(\sim p \vee q)$ are Logically Equivalent Sol: $\sim (p \wedge q) \rightarrow (\sim p \vee (\sim p \vee q))$ $\Rightarrow (\sim p \vee \sim q) \rightarrow (\sim p \vee (\sim p \vee q))$ $[\because \text{De Morgan's Law}]$ $\Rightarrow (\sim p \vee \sim q) \rightarrow (\sim p \vee \sim p \vee q)$ $[\because \text{Associative Law}]$ $\Rightarrow (\sim p \vee \sim q) \rightarrow (\sim p \vee q)$ $[\because \text{Idempotent Law}]$ $\Rightarrow \sim (\sim p \vee \sim q) \vee (\sim p \vee q)$ $[\because \text{Implication Law}]$ $\Rightarrow (\sim (\sim p) \wedge \sim (\sim q)) \vee (\sim p \vee q)$ $[\because \text{De Morgan's Law}]$ $\Rightarrow (p \wedge q) \vee (\sim p \vee q)$ $[\because \text{Double Negation Law}]$ $\Rightarrow p \vee (\sim p \vee q) \wedge q \vee (\sim p \vee q)$ $[\because \text{Distributive Law}]$ $\Rightarrow ((p \vee \sim p) \vee q) \wedge (q \vee \sim p \vee p)$ $[\because \text{Negation Law}]$ $\Rightarrow (T \vee q) \wedge (q \vee \sim p)$ $[\because \text{Idempotent Law}]$ $\Rightarrow (T \vee q) \wedge (q \vee \sim p)$ $[\because \text{Dominant Law}]$ $\Rightarrow T \wedge (\sim p \vee q)$ $[\because \text{Identity Law}]$ $\Rightarrow T \wedge (\sim p \vee q)$ $\Rightarrow (\sim p \vee q).$	https://youtu.be/fVwev765AOQ?s=i=Y5Lov4_EWlleCCdZ																												
6	Obtain the PDNF of negation of $p \vee (\neg p \wedge \neg q \wedge r)$. $\Rightarrow \sim p \wedge \sim (\sim p \wedge \sim q \wedge r)$ $[\because \text{De Morgan law}]$ $\Rightarrow \sim p \wedge (p \vee q \vee \sim r)$ $\Rightarrow (\sim p \wedge p) \vee (\sim p \wedge q) \vee (\sim p \wedge \sim r)$ $[\because \text{Distributive Law}]$ $\Rightarrow F \vee (\sim p \wedge q) \vee (\sim p \wedge \sim r)$ $\Rightarrow (\sim p \wedge q) \vee (\sim p \wedge \sim r)$ $[\because \text{Distributive Law}]$ $\Rightarrow [(\sim p \wedge q) \wedge r] \vee [(\sim p \wedge q) \wedge \sim r] \vee [(\sim p \wedge \sim r) \wedge q] \vee [(\sim p \wedge \sim r) \wedge \sim q]$ which is required PDNF.	https://youtu.be/zItFroaSmR0?si=J3k30U47vfMHWHQ0																												
7	If there was a meeting then catching the bus was difficult .If they arrived on time then catching the bus was not difficult .They arrived on time therefore there was no meeting .Show that these statements constitutes a valid argument P: There was a meeting q: Catching the bus was difficult r: They arrive on time. $p \rightarrow q, r \rightarrow \sim q, r \Rightarrow \sim p$ <table><tr><td>{1}</td><td>(1)</td><td>r</td><td>Rule P</td></tr><tr><td>{2}</td><td>(2)</td><td>$r \rightarrow \sim q$</td><td>Rule P</td></tr><tr><td>{1,2}</td><td>(3)</td><td>$\sim q$</td><td>Rule T, (1),(2)</td></tr><tr><td></td><td></td><td></td><td>Modulo ponens</td></tr><tr><td>{4}</td><td>(4)</td><td>$p \rightarrow q$</td><td>Rule P</td></tr><tr><td>{1,2,4}</td><td>(5)</td><td>$\sim p$</td><td>Rule T, (3),(4)</td></tr><tr><td></td><td></td><td></td><td>Modulo Tolens Rule.</td></tr></table> These arguments are valid.	{1}	(1)	r	Rule P	{2}	(2)	$r \rightarrow \sim q$	Rule P	{1,2}	(3)	$\sim q$	Rule T, (1),(2)				Modulo ponens	{4}	(4)	$p \rightarrow q$	Rule P	{1,2,4}	(5)	$\sim p$	Rule T, (3),(4)				Modulo Tolens Rule.	https://youtu.be/0AQYvo3Z0oY?s=i=_G76vq1MUaG9t0Lx
{1}	(1)	r	Rule P																											
{2}	(2)	$r \rightarrow \sim q$	Rule P																											
{1,2}	(3)	$\sim q$	Rule T, (1),(2)																											
			Modulo ponens																											
{4}	(4)	$p \rightarrow q$	Rule P																											
{1,2,4}	(5)	$\sim p$	Rule T, (3),(4)																											
			Modulo Tolens Rule.																											
8	Using the indirect method ,show that $p \rightarrow q, q \rightarrow r, \sim (p \wedge r), p \vee r$ is tautological implied with r	https://youtu.be/EIkKp-uz8ZU?si=tP3To05sLFSrSI7b																												

We prove that 'n' is valid.
By the method of proof of contradiction (or)
Indirect method of proof. Suppose 'n' is
invalid.

Therefore, n is valid. Now we prove that
 $P \rightarrow Q, Q \rightarrow n, n(P \wedge n), P \vee n, n$ are inconsistent.

{1}	(1)	$n n$	Rule P
{2}	(2)	$P \vee n$	Rule P
{1,2}	(3)	P	Rule T, (1), (2)
			disjunctive syllogism
{4}	(4)	$P \rightarrow Q$	Rule P
{1,2,4}	(5)	Q	Rule T, (3), (4)
			Modulo ponens
{6}	(6)	$Q \rightarrow n$	Rule P
{1,2,4,6}	(7)	n	Rule T, (5), (6)
			Modulo ponens
{1,2,4,6}	(8)	$n \wedge n$	Rule T, (1), (7)
		$\equiv F$	(P, $Q \rightarrow P \wedge Q$)
{9}	(9)	$n(P \wedge n)$	Rule P
{1,2,4,6,9}	(10)	$F \wedge (P \wedge n)$	Rule T, (8), (9)
			(P, $Q \rightarrow P \wedge Q$)
{1,2,4,6,9}	(11)	F	Rule T, (10)
			Dominant Law
all the premises are inconsistent our assumption 'n' is invalid is wrong.			
∴ 'n' is valid.			

- 9 a) By using CP rule, show that the following argument is valid.
"If A works hard then either B or C will enjoy. If B enjoys then A will not work hard. If D enjoys then C will not. Therefore, if A works hard then D will not enjoy".

Let P: A works hard			
Q: B will enjoy			
n: C will enjoy			
S: D will enjoy			
We prove that $P \rightarrow (Q \vee n), Q \rightarrow nP, S \rightarrow n n. \Rightarrow P \rightarrow nS$			
{1}	(1)	P	Rule P (additional Premise)
{2}	(2)	$P \rightarrow (Q \vee n)$	Rule P
{1,2}	(3)	$Q \vee n$	Rule T, (1), (2)
			Modulo ponens
{4}	(4)	$Q \rightarrow nP$	Rule P
{4}	(5)	$P \rightarrow nQ$	Rule T, (4)
			($P \rightarrow Q \equiv nQ \rightarrow nP$)
{1,4}	(6)	nQ	Rule T, (1), (5)
			Modulo ponens
{1,2,4}	(7)	n	Rule T, (3), (6)
			(disjunctive syllogism)
{8}	(8)	$S \rightarrow n n$	Rule P
{1,2,4,8}	(9)	nS	Rule T, (7), (8)
			Modulo Tollens
{1,2,4,8}	(10)	$P \rightarrow nS$	Rule CP

- b) Construct the truth table for $\sim (P \vee (Q \wedge R)) \leftrightarrow (P \vee Q) \wedge (P \vee R)$

- a) <https://youtu.be/Dafuq3qR>
<https://youtu.be/fVwev765AOQ?s>
[i=8ggr-diFGy8SoqNO](https://youtu.be/fVwev765AOQ?s)

10	a) Verify the validity of the following argument “ Every living thing is a plant or an animal. David’s dog is alive and it is not a plant. All animals have hearts. Hence, David’s dog has a heart.” b) Show that $R \wedge (P \vee Q)$ is a valid conclusion from the premises $P \vee Q, Q \rightarrow R, P \rightarrow M$ and $\sim M$.	a) https://youtu.be/-_OiYxHkhKM?si=VaDmZjdeGem3fWWv https://youtu.be/vjfxvsEAo_E?si=kAj0Q8K826DboHKZ
11	a) Show that $(\exists x)[P(x) \wedge Q(x)] \Rightarrow (\exists x)P(x) \wedge (\exists x)Q(x)$. b)	a) https://youtu.be/GP6e-HZhNNY?si=XpIjkTrpJCRpddb8 b) https://youtu.be/kNwDYv7ER5E?si=Ezehs3SOhEa76pkD

$\{1\}$	(1)	$(\exists x) (Px) \wedge Q(x)$	Rule P
$\{1\}$	(2)	$P(y) \wedge Q(y)$	Rule ES, (1)
$\{1\}$	(3)	$P(y)$	Rule T, (2)
$\{1\}$	(4)	$Q(y)$	Conjunctive Simplification

$\{1\}$	(5)	$(\exists x) P(x)$	Rule EG, (3)
$\{1\}$	(6)	$(\exists x) Q(x)$	Rule EG, (4)
$\{1\}$	(7)	$(\exists x) Px \wedge (\exists x) Q(x)$	Rule T, (5), (6) Conjunctive Simplification

b) Show that the premises $a \rightarrow (b \rightarrow c)$, $d \rightarrow (b \wedge \sim c)$, $a \wedge d$ are inconsistent

$\{1\}$	(1)	a	Rule P
$\{1\}$	(2)	a	Rule T, (1)
$\{1\}$	(3)	d	Rule T, (1)
$\{4\}$	(4)	$a \rightarrow (b \rightarrow c)$	Rule P
$\{1, 4\}$	(5)	$b \rightarrow c$	Rule T, (2), (4) Modulo Ponens
$\{1, 4\}$	(6)	$\sim b \vee c$	Rule T, (5) $(P \rightarrow Q \equiv \sim P \vee Q)$
$\{7\}$	(7)	$d \rightarrow (b \wedge \sim c)$	Rule P
$\{1, 7\}$	(8)	$b \wedge \sim c$	Rule T, (3), (7) Modulo Ponens
$\{1, 7\}$	(9)	$\sim(\sim b \vee c)$	Rule T, (8) De-morgan's law
$\{1, 4, 7\}$	(10)	$(\sim b \vee c) \wedge (\sim(\sim b \vee c))$	Rule T, (6), (9)
$\{1, 4, 7\}$	(11)	F	Rule T, (10) Negation law

12 a) Verify the validity of the following arguments:
All men are mortal.
Socrates is a man.
Therefore, Socrates is mortal

a) <https://youtu.be/ZpQpeBlCNTY?si=-iIlgds6vQKtnc2p>
<https://youtu.be/9Tg3jUheyEw?si=ivjtOVyesjHlbdJg>

Let $M(x)$: x is men
 $L(x)$: x is mortal
 s : Socrates

Symbolic representations above arguments is

$$\frac{(\forall x) (M(x) \rightarrow L(x))}{M(s)} \therefore L(s)$$

$\{1\}$ (1) $(\forall x) (M(x) \rightarrow L(x))$ Rule P
 $\{1\}$ (2) $M(s) \rightarrow L(s)$ Rule US, (1)
 $\{3\}$ (3) $M(s)$ Rule P
 $\{1, 3\}$ (4) $L(s)$ Rule T, (2), (3)
 Modulo Ponens

The given arguments are valid.

b) Obtain the PCNF of $(\neg p \rightarrow r) \wedge (q \leftrightarrow p)$. without using truth table

Soln: without using truth table.

$$\begin{aligned} & (\neg p \rightarrow r) \wedge (q \leftrightarrow p) \\ & \neg(\neg p \vee r) \wedge (q \leftrightarrow p) \quad [\because \text{Implication Law}] \\ & \equiv (p \vee r) \wedge (q \rightarrow p) \wedge (p \rightarrow q) \quad [\text{Double negation}] \\ & \equiv (p \vee r) \wedge (\neg q \vee p) \wedge (\neg p \vee q) \\ & \equiv (p \vee r \vee F) \wedge (\neg q \vee p \vee F) \wedge (\neg p \vee q \vee F) \quad [\because \text{Identity Law}] \\ & \equiv ((p \vee r) \vee (q \wedge \neg q)) \wedge ((\neg q \vee p) \vee (r \wedge \neg r)) \quad \begin{matrix} p \wedge T = p \\ p \vee F = p \end{matrix} \\ & \quad \wedge ((\neg p \vee q) \vee (r \wedge \neg r)) \quad [q \wedge \neg q = F] \\ & \equiv (p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee \neg q \vee r) \\ & \quad \wedge (\neg p \vee q \vee r) \wedge (\neg p \vee q \vee r) \\ & \equiv (p \vee q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee \neg q \vee r) \wedge \\ & \quad (\neg p \vee q \vee r) \wedge (\neg p \vee q \vee r) \end{aligned}$$

UNIT 2

Relations & Algebraic Structures: Relations, Properties of binary relations in a set, Relation Matrix and Graph of a Relation, Partition and covering of a set, equivalence relations, compatibility relations, composition of binary relations, Partial order relation, partially ordered set, Hasse diagram.

Algebraic structure, group, Abelian group, subgroup definitions and examples

1

Let $S = \{(1, 2), (1, 3), (2, 1), (3, 3)\}$ find S^2 .

Let $S = \{(1, 2), (1, 3), (2, 1), (3, 3)\}$. Find S^2 (the composition $S \circ S$).

By definition $S^2 = \{(a, c) \mid \exists b \text{ with } (a, b) \in S \text{ and } (b, c) \in S\}$. We check all possible intermediate b :

- From $(1, 2) \in S$: take $b = 2$. Pairs in S with first coordinate 2 are $(2, 1)$, so $(1, 1) \in S^2$.
- From $(1, 3) \in S$: take $b = 3$. Pairs in S with first coordinate 3 are $(3, 3)$, so $(1, 3) \in S^2$.
- From $(2, 1) \in S$: take $b = 1$. Pairs in S with first coordinate 1 are $(1, 2), (1, 3)$, so $(2, 2), (2, 3) \in S^2$.
- From $(3, 3) \in S$: take $b = 3$. Pairs in S with first coordinate 3 are $(3, 3)$, so $(3, 3) \in S^2$.

Hence

$$S^2 = \{(1, 1), (1, 3), (2, 2), (2, 3), (3, 3)\}.$$

<https://youtu.be/Yh-PbIKJS-w?si=Pd4chbOP7Ggy1QoI>

2

Let the set $A = \{1, 2, 3, 4\}$ and the relation $R = \{(1, 1), (1, 4), (4, 1), (4, 4), (2, 2), (2, 3), (3, 2), (3, 3)\}$ defined on A . Draw its matrix and graph.

https://youtu.be/EQBUDPIYeNw?si=2gUEvd_8e_qTUFdj

	Matrix of R Index rows and columns in order 1, 2, 3, 4. The adjacency matrix M_R has entry $m_{ij} = 1$ if $(i, j) \in R$, otherwise 0. $M_R = \begin{array}{c cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 1 & 0 & 0 & 1 \\ 2 & 0 & 1 & 1 & 0 \\ 3 & 0 & 1 & 1 & 0 \\ 4 & 1 & 0 & 0 & 1 \end{array}$ 1 Graph of R (directed)	
3	Define Monoid. Define Monoid. A <i>monoid</i> is an algebraic structure $(M, \cdot)$ consisting of a set M together with a binary operation $\cdot : M \times M \rightarrow M$ such that: (Associativity) For all $a, b, c \in M$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$. (Identity element) There exists an element $e \in M$ such that for all $a \in M$, $e \cdot a = a = a \cdot e$. Example: $(\mathbb{N}_0, +)$ (natural numbers including 0, with addition) is a monoid with identity 0.	https://youtu.be/dhriBg6XEro?si=jOAWvcnIs00zvIWV
4	Define group with respect to any binary operation. A <i>group</i> is a set G together with a binary operation $* : G \times G \rightarrow G$ satisfying: (Associativity) $\forall a, b, c \in G : (a * b) * c = a * (b * c)$. (Identity) $\exists e \in G$ such that $\forall a \in G : e * a = a = a * e$. (Inverse) $\forall a \in G$ there exists $a^{-1} \in G$ with $a * a^{-1} = e = a^{-1} * a$. If the operation is commutative ($a * b = b * a$ for all a, b), the group is called <i>abelian</i>.	https://youtu.be/dhriBg6XEro?si=jOAWvcnIs00zvIWV
5	If R and S are equivalence relations on set A. prove that $R \cap S$ is an equivalence relation. <i>Proof.</i> We show $R \cap S$ is reflexive, symmetric, transitive. Reflexive. Since R and S are equivalence relations, $\forall a \in A$ we have $(a, a) \in R$ and $(a, a) \in S$. Hence $(a, a) \in R \cap S$. So $R \cap S$ is reflexive. Symmetric. Suppose $(a, b) \in R \cap S$. Then $(a, b) \in R$ and $(a, b) \in S$. Because R, S are symmetric, $(b, a) \in R$ and $(b, a) \in S$. Therefore $(b, a) \in R \cap S$. Transitive. Suppose $(a, b) \in R \cap S$ and $(b, c) \in R \cap S$. Then $(a, b), (b, c) \in R$ and also $\in S$. Since R is transitive, $(a, c) \in R$. Since S is transitive, $(a, c) \in S$. Thus $(a, c) \in R \cap S$. Having all three properties, $R \cap S$ is an equivalence relation.	https://youtu.be/WoAGbTfP2eU?si=ge5rsyk6vnJVfpDa
6	Verify the following relation R on $X = \{1, 2, 3, 4\}$ is an equivalence relation or not? Given $R = \{(1, 1), (1, 4), (4, 1), (2, 2), (2, 3), (3, 4), (3, 3), (3, 2), (4, 3), (4, 4)\}$. Check properties: • Reflexive: $(1, 1), (2, 2), (3, 3), (4, 4)$ are all present, so reflexive holds. • Symmetric: For each pair: – $(1, 4)$ and $(4, 1)$ both present. – $(2, 3)$ and $(3, 2)$ both present. – $(3, 4)$ and $(4, 3)$ both present. So symmetry holds. • Transitive: need for all $(a, b), (b, c) \in R$ that $(a, c) \in R$. Consider $(1, 4) \in R$ and $(4, 3) \in R$. If transitive, we should have $(1, 3) \in R$. But $(1, 3) \notin R$. Thus transitivity fails. Therefore R is not an equivalence relation (counterexample: $(1, 4), (4, 3) \in R$ but $(1, 3) \notin R$).	https://youtu.be/Q5voInZi07c?si=WChRKluP_gF7GkZW
7	Show that the “greater than or equal” relation ($\geq$) is a partial ordering on the set of integers.	https://youtu.be/_MvUUNve6l4?s_i=AsWdb-HPJm7tY_Yx

	<i>Proof.</i> We check reflexivity, antisymmetry, transitivity on $\mathbb{Z}$. • Reflexive: For any $a \in \mathbb{Z}$, $a \geq a$ holds. • Antisymmetric: If $a \geq b$ and $b \geq a$, then $a = b$ (true for integers). • Transitive: If $a \geq b$ and $b \geq c$ then $a \geq c$. Hence $(\mathbb{Z}, \geq)$ is a partial order.	
8	Let $X = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{(x,y) \mid x-y \text{ is divisible by } 3\}$ in X. show that R is and Equivalence Relation. Answer : Reflexive: For any x, $x - x = 0$ divisible by 3, so $(x, x) \in R$. Symmetric: If $x - y$ is divisible by 3, then $y - x = -(x - y)$ is also divisible by 3, so $(y, x) \in R$. Transitive: If $x - y \equiv 0 \pmod{3}$ and $y - z \equiv 0 \pmod{3}$, then $x - z \equiv (x - y) + (y - z) \equiv 0 \pmod{3}$, so $(x, z) \in R$. Thus R is an equivalence relation. The equivalence classes (modulo 3) intersected with X are: $[1] = \{1, 4, 7\}$, $[2] = \{2, 5\}$, $[3] = \{3, 6\}$.	https://youtu.be/FXDITeHn568?si=rHK_S4GFQjHwD86v
9	(a) Define partition of a set and covering of sets, let $A1 = \{\{a,b\}, \{b,c\}\}$, $A2 = \{\{a\}, \{a,c\}\}$, $A3 = \{\{a\}, \{b,c\}\}$, $A4 = \{\{a,b,c\}\}$, $A5 = \{\{a\}, \{b\}, \{c\}\}$ and $A6 = \{\{a\}, \{a, b\}, \{a, c\}\}$. which of the following sets are partitions and covers the set $A = \{a, b, c\}$.	a) https://youtu.be/lelZcXUWJII?si=uVcywWU4kfRt0qAX https://youtu.be/f7cgCF4CqvY?si=IFQwtGO7jeDal1Gd

Question (verbatim):

Define partition of a set and covering of sets, let

$$A_1 = \{\{a, b\}, \{b, c\}\}, \quad A_2 = \{\{a\}, \{a, c\}\}, \quad A_3 = \{\{a\}, \{b, c\}\},$$

$$A_4 = \{\{a, b, c\}\}, \quad A_5 = \{\{a\}, \{b\}, \{c\}\}, \quad A_6 = \{\{a\}, \{a, b\}, \{a, c\}\}.$$

Which of the following sets are partitions and covers the set $A = \{a, b, c\}$.

4

Answer:

Definition (Covering of a Set). Let A be a non-empty set and let $\mathcal{C} = \{C_i \mid i \in I\}$ be a collection of non-empty subsets of A , where I is an index set. We say that $\mathcal{C}$ is a *covering* (or *cover*) of A if

$$\bigcup_{i \in I} C_i = A.$$

In other words, every element of A belongs to at least one subset in the collection $\mathcal{C}$. Note that in a covering, subsets may overlap (i.e., they need not be disjoint).

Definition (Partition of a Set). Let A be a non-empty set and let $\mathcal{P} = \{P_i \mid i \in I\}$ be a collection of non-empty subsets of A . We say that $\mathcal{P}$ is a *partition* of A if the following two conditions are satisfied:

1. **Covering condition:**

$$\bigcup_{i \in I} P_i = A.$$

That is, every element of A belongs to at least one subset P_i .

2. **Mutual disjointness:** For all $i, j \in I$ with $i \neq j$, we have

$$P_i \cap P_j = \emptyset.$$

That is, no element of A appears in more than one subset.

1. $A_1 = \{\{a, b\}, \{b, c\}\}.$

- Union: $\{a, b\} \cup \{b, c\} = \{a, b, c\} = A$. Hence A_1 is a **cover**.
- Intersection: $\{a, b\} \cap \{b, c\} = \{b\} \neq \emptyset$. So blocks are not pairwise disjoint; A_1 is **not a partition**.

2. $A_2 = \{\{a\}, \{a, c\}\}.$

- Union: $\{a\} \cup \{a, c\} = \{a, c\}$ which does not include b . So A_2 is **not a cover** and thus **not a partition**.

3. $A_3 = \{\{a\}, \{b, c\}\}.$

- Union: $\{a\} \cup \{b, c\} = \{a, b, c\} = A$. So A_3 is a **cover**.
- Intersection: $\{a\} \cap \{b, c\} = \emptyset$ and both blocks are nonempty. Therefore A_3 is a **partition**.

4. $A_4 = \{\{a, b, c\}\}.$

- Union: $\{a, b, c\} = A$ so A_4 is a **cover**.

5

- There is only one (nonempty) block, so pairwise disjointness holds vacuously; A_4 is a **partition** (the trivial partition).

5. $A_5 = \{\{a\}, \{b\}, \{c\}\}.$

- Union: $\{a\} \cup \{b\} \cup \{c\} = A$, so A_5 is a **cover**.
- The three sets are pairwise disjoint and nonempty; A_5 is a **partition**.

6. $A_6 = \{\{a\}, \{a, b\}, \{a, c\}\}.$

- Union: $\{a\} \cup \{a, b\} \cup \{a, c\} = \{a, b, c\} = A$, so A_6 is a **cover**.
- Blocks overlap (for example $a \in \{a\} \cap \{a, b\}$), therefore A_6 is **not a partition**.

Summary table:

Collection	Is a cover?	Is a partition?
A_1	Yes	No
A_2	No	No
A_3	Yes	Yes
A_4	Yes	Yes
A_5	Yes	Yes
A_6	Yes	No

(b) Determine whether the binary operation $*$ is commutative and associative on the set

(i) $\mathbb{Z}^+$, where $a*b = a+b+2$.

(ii) $\mathbb{Z}$, where $a*b = ab$.

Answer (i)

Commutativity: For all $a, b \in \mathbb{Z}_4$,

$$a * b = a + b + 2 = b + a + 2 = b * a,$$

so $*$ is commutative.

Associativity: Compute

$$(a * b) * c = (a + b + 2) + c + 2 = a + b + c + 4,$$

and

$$a * (b * c) = a + (b + c + 2) + 2 = a + b + c + 4.$$

Hence $(a * b) * c = a * (b * c)$ for all a, b, c , so $*$ is associative.

Remark: Although associative and commutative, $\mathbb{Z}_4$ with this operation does not have an identity in $\mathbb{Z}_4$ because solving $a * e = a$ gives $e = -2 \notin \mathbb{Z}_4$.

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Answer (ii)

On $\mathbb{Z}$ define $a * b = ab$ (ordinary integer multiplication).

Commutativity: For all $a, b \in \mathbb{Z}$, $ab = ba$, so $*$ is commutative.

Associativity: Integer multiplication is associative: $(ab)c = a(bc)$, so $*$ is associative.

10 (a) Let $A = \{2, 3, 6, 12, 24, 36\}$ and the relation on A defined as $a \leq b$ if a divides b . Draw the Hasse diagram for $(A, \leq)$.

Answer

We first determine the cover relations (i.e. $x < y$ when $x \mid y$ and there is no $z \in A$ with $x \mid z \mid y$ and $z \neq x, y$).

Divisibility relations among elements:

$$2 \mid 6, 2 \mid 12, 2 \mid 24, 2 \mid 36; \quad 3 \mid 6, 3 \mid 12, 3 \mid 24, 3 \mid 36;$$

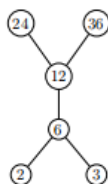
$$6 \mid 12, 6 \mid 24, 6 \mid 36; \quad 12 \mid 24, 12 \mid 36.$$

Covers (no intermediate in A):

$$2 < 6, \quad 3 < 6, \quad 6 < 12, \quad 12 < 24, \quad 12 < 36.$$

Note that 24 and 36 are distinct maximal elements and $24 \nmid 36, 36 \nmid 24$.

Hasse diagram (levels bottom to top): bottom $\{2, 3\}$, then $\{6\}$, then $\{12\}$, top $\{24, 36\}$



(b) Show that $(\mathbb{Z}_5 - \{0\}, \times_5)$ is an abelian group

Answer

Let $G = \{1, 2, 3, 4\}$ be the nonzero residues modulo 5, with operation multiplication modulo 5 (denote $\times_5$).

7

• Closure: If $a, b \in G$, then $ab \not\equiv 0 \pmod{5}$ (since 5 is prime), so $ab \pmod{5} \in G$.

• Associativity: Follows from associativity of integer multiplication; reduction mod 5 respects the product.

• Identity: 1 is the identity: $1 \cdot a \equiv a \pmod{5}$.

• Inverses: Each element has a multiplicative inverse mod 5:

$$1^{-1} = 1, \quad 2^{-1} = 3 \text{ (since } 2 \cdot 3 = 6 \equiv 1), \quad 3^{-1} = 2, \quad 4^{-1} = 4 \text{ (since } 4 \cdot 4 = 16 \equiv 1).$$

• Commutativity: Multiplication mod 5 is commutative.

Thus $(\mathbb{Z}_5^\times, \times_5)$ is an abelian group.

A) <https://youtu.be/GpBfvE5zFJY?si=05qoZJqljsZCPTUS>
https://youtu.be/6_KJwRLegB4?s_i=w-b0hClnGdkUGaWC

11 (a) Show that $(\mathbb{Z}, *)$ is an abelian group with respect to the binary operation $*$, which is defined by $a * b = a + b + 1$.

A) <https://youtu.be/jud2EpRPWT8?si=TXNZzCOSqFBC-Oi7>
https://youtu.be/DW5vO7hdsQs?s_i=yhRBw45z07l_GHnF

Answer

Let $*$ be defined on $\mathbb{Z}$ by $a * b = a + b + 1$.

- **Closure:** $a + b + 1 \in \mathbb{Z}$ for all $a, b \in \mathbb{Z}$.
- **Associativity:** For all $a, b, c \in \mathbb{Z}$,
 $(a * b) * c = (a + b + 1) + c + 1 = a + b + c + 2 = a + (b + c + 1) + 1 = a * (b * c)$.
- **Identity:** Find $e \in \mathbb{Z}$ with $a * e = a$ for all a . Solve $a + e + 1 = a$, so $e = -1$. Check:
 $a * (-1) = a + (-1) + 1 = a$.
- **Inverses:** For given a , solve $a * b = e = -1$. That gives $a + b + 1 = -1 \Rightarrow b = -a - 2 \in \mathbb{Z}$.
 Thus every a has inverse $-a - 2$.
- **Commutativity:** $a * b = a + b + 1 = b + a + 1 = b * a$.

Therefore $(\mathbb{Z}, *)$ is an abelian group (identity -1 , inverse of a is $-a - 2$).

(b) Give $S = \{1, 2, 3, 4, \dots, 10\}$ and relation R on S where $R = \{(x, y) / x + y = 10\}$ what are the properties of the relation.

Answer

List the pairs in R explicitly:

$$R = \{(1, 9), (2, 8), (3, 7), (4, 6), (5, 5), (6, 4), (7, 3), (8, 2), (9, 1)\}.$$

Check relation properties on S :

- **Reflexive:** A relation is reflexive if $(x, x) \in R$ for all $x \in S$. Here only $(5, 5) \in R$. Hence R is *not reflexive*.
- **Symmetric:** If $(x, y) \in R$ then $x + y = 10$ implies $y + x = 10$, so $(y, x) \in R$. Thus R is *symmetric*.
- **Transitive:** If $(x, y), (y, z) \in R$, then $x + y = 10$ and $y + z = 10$. Subtracting gives $x = z$. For (x, z) to be in R we need $x + z = 10$, hence $2x = 10$ which forces $x = 5$. So transitivity does not hold generally (only in trivial cases). Thus R is *not transitive*.
- **Antisymmetric:** There exist distinct x, y (e.g. $1, 9$) with $(1, 9), (9, 1) \in R$, so R is *not antisymmetric*.

Conclusion: R is *symmetric only* (neither reflexive, nor transitive, nor antisymmetric). It is not an equivalence relation.

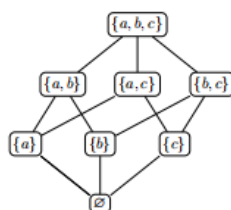
12 (a) Draw the Hasse Diagram for poset $\{(A, B) : A \subseteq B\}$ on the power set $P(S)$ where $S = \{a, b, c\}$ and $\subseteq$ is a relation.

Answer

The power set $P(S)$ has the eight elements

$$\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}.$$

The poset under inclusion is the Boolean lattice of rank 3. The Hasse diagram (cover relations are between sets that differ by exactly one element) is shown below.



(b) Let R and S be two relations on set of positive integers
 $R = \{(x, 2x) / x \in \mathbb{Z}^+\}$, $S = \{(x, 7x) / x \in \mathbb{Z}^+\}$ find $R \circ R$, $R \circ S$, $R \circ R \circ R$ and $R \circ S \circ R$.

A) https://youtu.be/GpBfvE5zFJY?si=L99vT6L_KTxnVvIk
<https://youtu.be/ABDWAGLRqo8?si=GFdZOhlUV52-V19F>

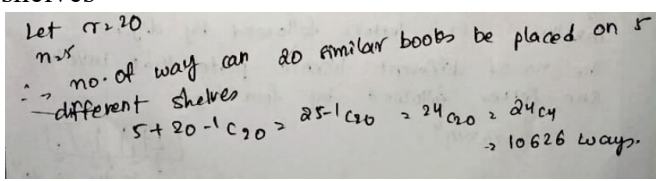
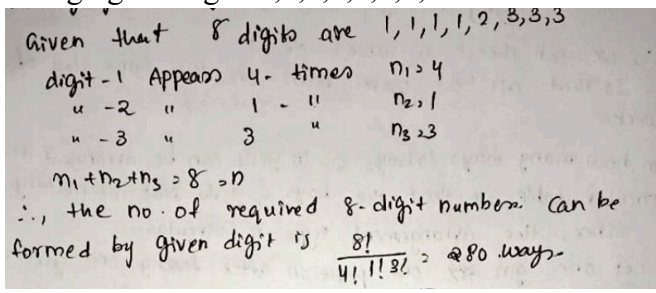
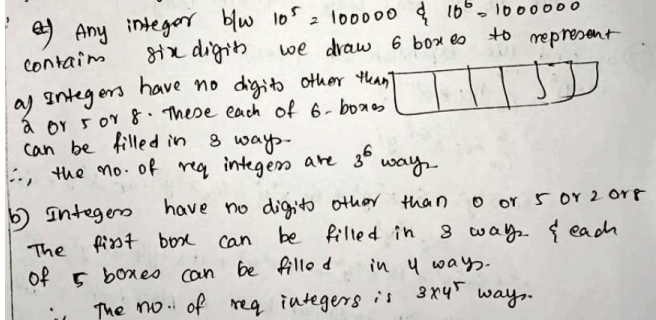
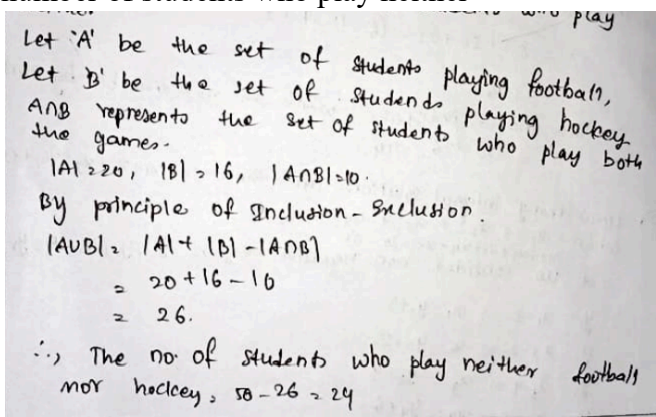
	Answer Interpret each relation as a function-like map: $R : x \mapsto 2x, \quad S : x \mapsto 7x.$ Composition of relations corresponds to composition of these linear maps. $R \circ R = \{(x, 2(2x)) \mid x \in \mathbb{Z}^+\} = \{(x, 4x)\}.$ $R \circ S = \{(x, 2(7x)) \mid x \in \mathbb{Z}^+\} = \{(x, 14x)\}.$ $R \circ R \circ R = \{(x, 2(2(2x))) \mid x \in \mathbb{Z}^+\} = \{(x, 8x)\}.$ $R \circ S \circ R = \{(x, 2(7(2x))) \mid x \in \mathbb{Z}^+\} = \{(x, 28x)\}.$ More generally, $R^n = \{(x, 2^n x)\}$ and compositions mixing R and S multiply by the corresponding constants in the order of composition.	
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UNIT 3

Elementary Combinatorics :

Basics of Counting, Combinations and Permutations, Enumeration of Combinations and Permutations, Enumerating Combinations and Permutations with Repetitions, Enumerating Permutations with Constrained Repetitions, Binomial Coefficients, The Binomial and Multinomial Theorems. The Principle of Inclusion-Exclusion, Pigeonhole principle.

1	In how ways 10 boys and 10 girls can be arranged in a circular table so that boys and girls seat alternatively Solution. Fix one boy to break rotational symmetry. The remaining 9 boys can be arranged in the 9 alternate boy seats in $9!$ ways. The 10 girls can be arranged in the 10 girl seats in $10!$ ways. Hence, $\boxed{9! \cdot 10!}.$	https://youtu.be/kxoh_4x3oAk?si=ll1Q1b-p40AjrmlQL
2	Find n if i) $P(n,2)=72$ Solution. $P(n,2) = n(n-1) = 72 \Rightarrow n^2 - n - 72 = 0.$ $n = \frac{1 \pm \sqrt{1+288}}{2} = \frac{1 \pm 17}{2} \Rightarrow n = 9 \text{ (positive root).}$ $\boxed{n = 9}.$	https://youtu.be/injbcHXCVE0?si=xDdfQT7PRCxxgV2cL
3	Find the number of arrangements of letters in the word MAYAJALAM? Solution. Letters: $A \times 4, M \times 2, Y \times 1, J \times 1, L \times 1$ (total 9). $\text{Arrangements} = \frac{9!}{4!2!} = \boxed{7560}.$	https://youtu.be/vmhCFr8Y-4A?si=E_M8eg24TUP9m6jd
4	State pigeonhole principle? Solution. If $n+1$ (or more) objects are placed into n boxes, then at least one box contains at least 2 objects. <i>Generalized form:</i> If N objects are placed into k boxes, then some box contains at least $\left\lceil \frac{N}{k} \right\rceil \text{ objects.}$	https://youtu.be/uP_b9wCq5LQ?si=mR7XsHPPsAaOaqam
5	a Find the expansion of $(x+y)^6$ <i>Find the Expansion of $(x+y)^6$.</i> <i>From the Binomial Theorem we have $\sum_{r=0}^n {}^n C_r x^{n-r} y^r$.</i> $(x+y)^6 = \sum_{r=0}^6 {}^6 C_r x^{6-r} y^r = {}^6 C_0 x^6 y^0 + {}^6 C_1 x^5 y^1 + {}^6 C_2 x^4 y^2 + {}^6 C_3 x^3 y^3 + {}^6 C_4 x^2 y^4 + {}^6 C_5 x y^5 + {}^6 C_6 x^0 y^6.$ $= x^6 + 6x^5y + 15x^4y^2 + 20x^3y^3 + 15x^2y^4 + 6xy^5 + y^6.$	a) https://youtu.be/7KtMzqWB9L8?si=vDBTrttTXZaMmMe2 b) https://youtu.be/fKAjPkdzCD4?si=d1o1xwIluHYWe900 c) https://youtu.be/ukW61smUeGU?si=GEKwjiGzj2n1jbiq

	b. How many ways can 20 similar books be placed on 5 different shelves  Let $n = 20$ $m = 5$ $\therefore$ no. of ways can 20 similar books be placed on 5 different shelves $5 + 20 - 1C_{20} = 24C_4 = 24C_4 = 10626$ ways.	
	b) How many different 8 digit numbers can be formed by arranging the digits 1,1,1,1,2,3,3,3?  Given that 8 digits are 1, 1, 1, 1, 2, 3, 3, 3 digit - 1 Appears 4 - times $n_1 = 4$ " - 2 " 1 - " $n_2 = 1$ " - 3 " 3 - " $n_3 = 3$ $n_1 + n_2 + n_3 = 8 = n$ $\therefore$ the no. of required 8-digit numbers can be formed by given digit is $\frac{8!}{4!1!3!} = 280$ ways.	
6	How many integers between 10^5 and 10^6 a) Have no digits other than 2, 5 or 8 b) Have no digits other than 0, 2, 5, or 8  a) Any integer b/w $10^5 = 100000$ & $10^6 = 1000000$ contains six digits. We draw 6 boxes to represent a) integers have no digits other than 2 or 5 or 8. These each of 6-boxes can be filled in 3 ways. $\therefore$ the no. of req integers are 3^6 ways. b) Integers have no digits other than 0 or 5 or 2 or 8. The first box can be filled in 3 ways & each of 5 boxes can be filled in 4 ways. $\therefore$ The no. of req integers is 3×4^5 ways.	https://youtu.be/fKAjPkdzCD4?si=d1o1xwI1uHYWe90O
7	In a class of 50 students 20 students play football and 16 students play hockey. It is found that 10 students play both the games. Find the number of students who play neither  Let 'A' be the set of students playing football, Let 'B' be the set of students playing hockey. $A \cap B$ represents the set of students who play both the games. $A = 20$, $B = 16$, $A \cap B = 10$. By principle of Inclusion - Exclusion. $A \cup B = A + B - A \cap B$ $= 20 + 16 - 10$ $= 26$. $\therefore$ The no. of students who play neither football nor hockey, $50 - 26 = 24$	https://youtu.be/UrvJBnl7E8k?si=UbwIFRLtproUdGQY
8	Prove that if any 30 people are selected, then we may choose a subset of 5 so that all 5 were born on the same day of the week Solution. There are 7 days of the week (boxes) and 30 people (objects). By the generalized pigeonhole principle, $\text{max load} \geq \left\lceil \frac{30}{7} \right\rceil = 5.$ Therefore, at least one day has ≥ 5 birthdays, yielding a subset of 5 with the same weekday of birth. $\square$	https://youtu.be/XGHkx6Qd174?si=JHPTOSO87VWfC1oS
9	(a) Find the coefficient of $u^2w^3x^4y^2$ in the expansion of $(u+v+2w+x+3y+z)^{11}$	A) https://youtu.be/0kvQkwtXSU?si=JGF4XzjHv-xPhT2v

	Solution. Use the multinomial theorem. We need the term with exponents $(u, v, w, x, y, z) = (2, 0, 3, 4, 2, 0), \quad \text{sum} = 11.$ The coefficient equals $\frac{11!}{2!0!3!4!2!0!} \cdot (1)^2(1)^0(2)^3(1)^4(3)^2(1)^0 = \frac{11!}{2!3!4!2!} \cdot 8 \cdot 9.$ Compute: $\frac{11!}{2!3!4!2!} = 69300 \Rightarrow [69300 \times 72 = 4,989,600].$ (b) How many bit strings of length 8 contain (i) exactly five 1s? (ii) an equal number of 0s and 1s? (iii) at least four 1s? (iv) at least three 1s and at least three 0s? Solution. Choose positions of the 1s. (i) $\binom{8}{5} = [56].$ (ii) $\binom{8}{4} = [70].$ (iii) $\sum_{k=4}^8 \binom{8}{k} = \binom{8}{4} + \binom{8}{5} + \binom{8}{6} + \binom{8}{7} + \binom{8}{8} = 70 + 56 + 28 + 8 + 1 = [163].$ (iv) "At least three 1s and at least three 0s" $\Leftrightarrow$ number of 1s $k \in \{3, 4, 5\}$: $\binom{8}{3} + \binom{8}{4} + \binom{8}{5} = 56 + 70 + 56 = [182].$	https://youtu.be/xuCVIS1V69A?si=9zo8vmoS6zCcvnpi
10	a) Consider the six digits 1,2,3,5,6 and 7. Assuming that repetitions are not permitted, answer the following: (i) How many four digit numbers can be formed from the six digits 1,2,3,5,6 and 7? (ii) How many of these numbers are less than 4000? (iii) How many of the numbers in (i) are even? (iv) How many of the numbers in (i) are odd? (v) How many of the numbers in (i) are multiple of 5? Solution. (i) $P(6, 4) = 6 \cdot 5 \cdot 4 \cdot 3 = [360].$ (ii) First digit $\in \{1, 2, 3\}$ (3 choices), then $P(5, 3) = 60$ for the rest: $3 \cdot 60 = [180].$ (iii) Last digit even $\in \{2, 6\}$ (2 choices), then $P(5, 3) = 60$: $2 \cdot 60 = [120].$ (iv) Either $360 - 120$ or last digit odd $\in \{1, 3, 5, 7\}$ (4 choices), then $P(5, 3) = 60$: $[240].$ (v) Last digit = 5 (forced), then $P(5, 3) = [60].$ b) Find the coefficient of $x^{101}y^{99}$ in $(2x - 3y)^{200}$? In $(2x - 3y)^{200}$, the general term with y^k is $\binom{200}{k}(2x)^{200-k}(-3y)^k$. For $x^{101}y^{99}$, take $k = 99$. $\binom{200}{99} 2^{101} (-3)^{99} = -\binom{200}{99} 2^{101} 3^{99}.$	A) https://youtu.be/9pTv3zSQd1A?si=efE-MMf4OxWNmAmI https://youtu.be/raks0IZ2Saw?si=SPtKnHLR3V8UXo-4
11	(a) Suppose a department consists of eight men and nine women. In how many ways can we select a committee of (i) three men and four women? (ii) four persons that has at least one woman?	A) https://youtu.be/2Ekqr98SvKQ?si=hyEdfaBAL-oBz66 https://youtu.be/KPDE6vRWE8I?si=YZQ-7xtwDkTdCsxo

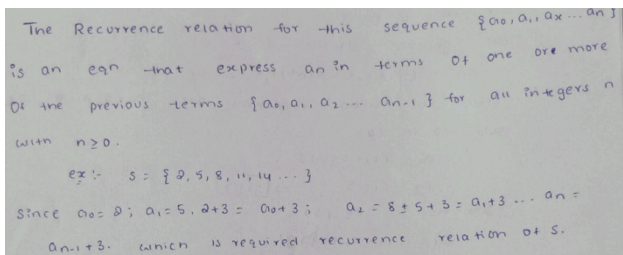
	(b.i) Choose men and women independently: $\binom{8}{3} \binom{9}{4} = 56 \cdot 126 = \boxed{7056}.$ (b.ii) Total 4-person committees minus all-men committees: $\binom{17}{4} - \binom{8}{4} = 2380 - 70 = \boxed{2310}.$ (b) How many integral solutions are there to $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ Where $x_1 \geq 3, x_2 \geq 2, x_3 \geq 4, x_4 \geq 6, x_5 \geq 0$. Let $y_1 = x_1 - 3, y_2 = x_2 - 2, y_3 = x_3 - 4, y_4 = x_4 - 6, y_5 = x_5$. Then $y_i \geq 0$ and $y_1 + y_2 + y_3 + y_4 + y_5 = 20 - (3 + 2 + 4 + 6) = 5.$ Number of nonnegative solutions: $\binom{5 + 5 - 1}{5 - 1} = \binom{9}{4} = \boxed{126}.$	
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12	a). Consider a set of integers from 1 to 250. Find how many of these numbers are not divisible by 3 or 5 or 7. Solution (Inclusion-Exclusion). Let A, B, C be multiples of 3, 5, 7 respectively. $ A = \left\lfloor \frac{250}{3} \right\rfloor = 83, \quad B = \left\lfloor \frac{250}{5} \right\rfloor = 50, \quad C = \left\lfloor \frac{250}{7} \right\rfloor = 35.$ Intersections: $ A \cap B = \left\lfloor \frac{250}{15} \right\rfloor = 16, \quad A \cap C = \left\lfloor \frac{250}{21} \right\rfloor = 11, \quad B \cap C = \left\lfloor \frac{250}{35} \right\rfloor = 7,$ $ A \cap B \cap C = \left\lfloor \frac{250}{105} \right\rfloor = 2.$ Thus, $A \cup B \cup C = 83 + 50 + 35 - (16 + 11 + 7) + 2 = 136$. Hence, not divisible by any of 3, 5, 7: $250 - 136 = \boxed{114}.$ b) How many different licence plates are there that involve 1, 2 or 3 letters followed by 4 digits Solution. Assume 26 letters (A-Z) and digits 0-9; repetitions allowed. For the letter part: $26 + 26^2 + 26^3$ choices. For the digit part: 10^4 choices. $(26 + 26^2 + 26^3) \cdot 10^4 = 18,278 \cdot 10^4 = 182,780,000.$	A) https://youtu.be/pThd71uovk0?si=8xDYjRunlvTLib7 B) https://youtu.be/UTRL3EEU6qE?si=YfDvBUmQJxC62Pu7
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UNIT 4

Recurrence Relations

Formulation as Recurrence Relations, Solving Recurrence Relations - Substitution and Method of Characteristic Roots, Method of Undetermined coefficients. Generating Functions: Generating Functions of Sequences, Calculating Coefficient of expansions.

1	Define recurrence relation. 	https://youtu.be/UwZCaQLjpYs?si=Nx8KT1VbhHxoxTSd
2	Define Fibonacci sequence and write it's recurrence relation	https://youtu.be/N7pc4aVBP5I?si=DDiSfCXnq9EKnkj

	The sequence $\{1, 1, 2, 3, 5, 8, 13, \dots\}$ is called Fibonacci series. The RR for the f.s is $f_n = f_{n-1} + f_{n-2}$ for $n \geq 2$ and where $f_0 = 1, f_1 = 1$ is called initial condition or basis.	
3	Find a particular solution for recurrence relation using the method of determined coefficients $a_n - 5a_{n-4} = 3^n$? Ans: To find P.T., R.H.S of (1), $1 \cdot 3^n$ where 3 is not characteristic root (since 3 is not satisfying the characteristic equation). $a_n = A_0 3^n \rightarrow (2)$ Consider $a_n - 5a_{n-4} = 3^n$. $A_0 3^n - 5A_0 3^{n-4} = 3^n$. $\div 3^n (A_0 - 5A_0 \cdot \frac{1}{81}) = 1$ $(\frac{3 \times 3 \times 3 \times 3}{81} = 81)$ $A_0 [1 - \frac{5}{81}] = 1$ $A_0 [\frac{81-5}{81}] = 1$ $A_0 [\frac{76}{81}] = 1$ $A_0 = \frac{81}{76}$ Sub A in (2), $a_n = \frac{81}{76} 3^n$ which is required particular solution.	https://youtu.be/x7gpcVGTIys?si=d7ill-Mx9BJNDKCH
4	Find the generating function of the sequence 1, 2, 3, 4, ...? $A(x) = 1 + 2x + 3x^2 + 4x^3 + \dots$ $= \sum_{n=0}^{\infty} (n+1)x^n = (1-x)^{-2}$ $\therefore$ the generating function for the sequence is $(1-x)^{-2}$	https://youtu.be/OcNtFcsCztA?si=EzvRrk55X1FGEisu
5	Solve the recurrence relation $a_{n+2} - 5a_{n+1} + 4a_n = 0$ for $n \geq 0$, given $a_0 = 0$, and $a_1 = 13$. Ans: (2) solve the recurrence relation $a_{n+2} - 5a_{n+1} + 4a_n = 0$ for given $a_0 = 0$ & $a_1 = 13$ The given characteristic eqⁿ $a_{n+2} - 5a_{n+1} + 4a_n = 0 \rightarrow (1)$ $m^2 - 5m + 4 = 0$ $m^2 - 4m - m + 4 = 0$ $m(m-4) - 1(m-4) = 0$ $(m-1)(m-4) = 0$ $m = 1, 4$ roots are real & distinct so, characteristic roots are $a_n = C_1(1)^n + C_2(4)^n \rightarrow (2)$ $a_0 = 0, a_1 = 13$ sub $n=0$ in eqⁿ (2) $a_0 = C_1(1)^0 + C_2(4)^0$ $0 = C_1 + C_2 \rightarrow (3)$ sub $n=1$ $a_1 = C_1(1)^1 + C_2(4)^1$ $13 = C_1 + 4C_2 \rightarrow (4)$ Solve (3) & (4) $C_1 + C_2 = 0$ $C_1 + 4C_2 = 13$ $-3C_2 = -13$ $C_2 = \frac{13}{3}$ sub C_2 in (2) $C_1 + \frac{13}{3} = 0$ $C_1 = -\frac{13}{3}$ sub C_1, C_2 in (2) $a_n = -\frac{13}{3}(1)^n + \frac{13}{3}(4)^n$	https://youtu.be/TkAS1RktMsU?s=FRhNQCUKSMLz7m9E
6	Solve the recurrence relation $a_n = a_{n-1} + 3n^2 + 3n + 1$ where $a_0 = 1$. Given that $a_0 = 1$ for $n=1, a_1 = a_0 + 3(1)^2 + 3(1) + 1$ $\therefore 1 + 3 + 3 + 1 = 8$ $a_1 = 8$ $n=2, a_2 = a_1 + 3(2)^2 + 3(2) + 1$ $= 8 + 12 + 6 + 1 = 27$ $a_2 = 27$ $n=3, a_3 = a_2 + 3(3)^2 + 3(3) + 1$ $= 27 + 27 + 9 + 1 = 64$ $a_3 = 64$ $\therefore$ $a_n = (n+1)^3$ this is the req^d solⁿ of given R.R.	https://youtu.be/D2t3R7YIomY?si=xmvp7PawP2QD1fyO

7

Find the unique solution to the recurrence relation with the given initial conditions:

$$a_n = 9a_{n-1} - 27a_{n-2} + 27a_{n-3} \text{ with } a_0 = 5, a_1 = 24, a_2 = 117$$

Sol Given that $a_n - 9a_{n-1} + 27a_{n-2} - 27a_{n-3} = 0$ for $n \geq 3$, which is a homogeneous linear recurrence relation of degree 3.

The characteristic equation is $C(m) = 0$.

$$m^3 - 9m^2 + 27m - 27 = 0. \quad (\text{Initial root } m=3)$$

By synthetic division method

$$m=3 \begin{array}{r|rrrr} 1 & -9 & 27 & -27 & 0 \\ 0 & 3 & -18 & 27 & \\ \hline 1 & -6 & 9 & 0 & \end{array}$$

$$\therefore m=3, m^2 - 6m + 9 = 0$$

$$m=3 \quad (m-3)^2 = 0$$

$$m=3, 3, 3$$

$\therefore$ The characteristic roots are $m=3, 3, 3$ which are real and equal.

$\therefore$ The general solution of the given recurrence relation is

$$a_n = C.F = (C_1 + C_2 n + C_3 n^2)(3)^n \rightarrow (1)$$

Given initial conditions are $a_0 = 5, a_1 = 24, a_2 = 117$.

$$\text{Sub } n=0 \text{ in (1), } a_0 = (C_1 + C_2(0) + C_3(0))3^0$$

$$\Rightarrow C_1 = 5 \rightarrow (2)$$

$$\text{Sub } n=1 \text{ in (1), } a_1 = (C_1 + C_2 + C_3)(3)^1$$

$$\Rightarrow C_1 + C_2 + C_3 = \frac{24}{3} = 8$$

$$\Rightarrow C_2 + C_3 = 8 - C_1 = 8 - 5 = 3$$

$$\Rightarrow C_2 + C_3 = 3 \rightarrow (3)$$

$$\text{Sub } n=2 \text{ in (1), } a_2 = (C_1 + C_2(2) + C_3(4))3^2$$

$$\Rightarrow 117 = (C_1 + 2C_2 + 4C_3)(9)$$

$$\Rightarrow C_1 + 2C_2 + 4C_3 = \frac{117}{9} = 13$$

$$\Rightarrow 2C_2 + 4C_3 = 13 - C_1 = 13 - 5 = 8$$

$$\therefore 2C_2 + 4C_3 = 8$$

$$C_2 + 2C_3 = 4 \rightarrow (4)$$

Subtract (3) & (4).

$$\begin{array}{r} C_2 + C_3 = 3 \\ C_2 + 2C_3 = 4 \\ \hline -C_3 = -1 \end{array}$$

$$C_3 = 1 \rightarrow (5)$$

$$\text{Sub } C_3 \text{ in (3), we get } C_2 = 3 - C_3 = 3 - 1 = 2 \Rightarrow C_2 = 2 \rightarrow (6)$$

Substitute C_1, C_2, C_3 in (1)

$$\therefore a_n = (5 + 2n + n^2)3^n$$

[https://youtu.be/KW7PQ4RCcLo?](https://youtu.be/KW7PQ4RCcLo?si=_cXWfDCNjiZRtDsz)

[si=_cXWfDCNjiZRtDsz](https://youtu.be/KW7PQ4RCcLo?si=_cXWfDCNjiZRtDsz)

8

Find a particular solution for the recurrence relation using the method of undetermined coefficients: $a_n - 3a_{n-1} + 2a_{n-2} = 5n + 3$.

<https://youtu.be/WVu9aCi7qDg?s>

[i=YCg0txPwHb5NhNuv](https://youtu.be/WVu9aCi7qDg?s)

Sol Given that $a_n - 3a_{n-1} + 2a_{n-2} = 5n + 3 \rightarrow (1)$
 To find C.F, consider $a_n - 3a_{n-1} + 2a_{n-2} = 0$ which is a homogeneous linear recurrence relation of degree 2.
 The complementary function is $CC(m) = 0$.
 $\Rightarrow m^2 - 3m + 2 = 0$
 $\Rightarrow m^2 - 2m - m + 2 = 0$
 $\Rightarrow m(m-2) - 1(m-2) = 0$
 $\Rightarrow (m-2)(m-1) = 0$
 $\Rightarrow m = 1, 2$.
 The characteristic roots are $m = 1, 2$ which are real and distinct.
 Complementary function of (1) is
 $a_n^h = C_1(1)^n + C_2(2)^n \rightarrow (2)$

To find P.I, Since R.H.S of eqⁿ (1) is $(5n+3) \cdot 1^n$ where 1 is not a characteristic root of multiplicity 1. (3)
 $a_n^p = (A_0 + A_1 n) \cdot 1^n \cdot n^1$ ($a_n^p = (A_0 + A_1 n + \dots + A_r n^r) \cdot 1^n \cdot n^m$)
 $\Rightarrow a_n^p = A_0 n + A_1 n^2 \rightarrow (3)$
 Substitute above values in (1).
 from (1) $a_n - 3a_{n-1} + 2a_{n-2} = 5n + 3$.
 $\Rightarrow (A_0 n + A_1 n^2) - 3(A_0(n-1) + A_1(n-1)^2) + 2(A_0(n-2) + A_1(n-2)^2) = 5n + 3$
 $= 5n + 3$
 $\Rightarrow A_0 n + A_1 n^2 - 3A_0 n + 3A_0 + 3A_1 n^2 - 6A_1 n + 3A_1 + 2A_0 n - 4A_0 + 2A_1 n^2 - 8A_1 n + 8A_1 = 5n + 3$
 $\Rightarrow (A_1 - 3A_1 + 2A_1)n^2 + (A_0 - 3A_0 + 2A_0 + 6A_1 - 8A_1)n + (3A_0 - 3A_1 - 4A_0 + 8A_1) = 5n + 3$
 $\Rightarrow -2A_1 n + (-A_0 + 5A_1) = 5n + 3$
 Compare coefficient of n on both sides $-2A_1 = 5 \Rightarrow A_1 = -5/2$
 Compare coefficient of constants O.B.S. $-A_0 + 5A_1 = 3$
 $-A_0 - 25/2 = 3 \Rightarrow A_0 = -31/2$
 Substitute the values of A_0, A_1 in (3) we get $a_n^p = -31/2 n - 5/2 n^2$

$\therefore$ the general solution of given recurrence relation is
 $a_n = a_n^h + a_n^p$ (from (2) & (4) sub a_n^h & a_n^p).
 $a_n = C_1(1)^n + C_2(2)^n - \frac{31}{2}n - \frac{5}{2}n^2$

9 Solve the recurrence relation
 $a_k - 7a_{k-1} + 10a_{k-2} = 6 + 8k, a(0) = 1, a(1) = 2$

https://youtu.be/DN7rZ2R4Eok?si=XHJxfQWfqgUqF9_K

Sol Given that $a_k - 7a_{k-1} + 10a_{k-2} = 6 + 8k \rightarrow (1)$.
 To find C.F, Consider $a_k - 7a_{k-1} + 10a_{k-2} = 0$ is a homogeneous linear recurrence relation of degree 2.
 The characteristic equation is $C(m) = 0$.
 $m^2 - 7m + 10 = 0$.
 $\Rightarrow m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{7 \pm \sqrt{49 - 4(10)}}{2}$
 $= \frac{7 \pm \sqrt{49 - 40}}{2} = \frac{7 \pm \sqrt{9}}{2} = \frac{7 \pm 3}{2} = \frac{7+3}{2}, \frac{7-3}{2}$
 $= \frac{10}{2}, \frac{4}{2} = 5, 2$.
 $\therefore$ The characteristic roots are 2, 5 which are distinct and real.
 The complementary function of (1) is
 $a_k^h = C.F = C_1 2^k + C_2 5^k \rightarrow (2)$.
 To find P.S, R.H.S of (1) is $(6 + 8k) \cdot 1^k$ where 1 is not characteristic root.
 $a_k^p = (A_0 + A_1 k) \cdot 1^k$.
 $a_k^p = A_0 + A_1 k \rightarrow (3)$.

Consider $a_k - 7a_{k-1} + 10a_{k-2} = 6 + 8k$.
 $A_0 + A_1 k - 7(A_0 + A_1(k-1)) + 10(A_0 + A_1(k-2)) = 6 + 8k$.
 $\Rightarrow A_0 + A_1 k - 7A_0 - 7A_1 k + 7A_1 + 10A_0 + 10A_1 k - 20A_1 = 6 + 8k$.
 $\Rightarrow A_0 + A_1 k - 7A_0 - 7A_1 k + 7A_1 + 10A_0 + 10A_1 k - 20A_1 = 6 + 8k$.
 $\Rightarrow (A_1 - 7A_1 + 10A_1)k + (A_0 - 7A_0 + 7A_1 + 10A_0 - 20A_1) = 6 + 8k$.
 $\Rightarrow 4A_1 k + (4A_0 - 13A_1) = 8k + 6$.
 Compare coefficient of k on both sides. $4A_1 = 8 \Rightarrow A_1 = 2$.
 Compare coefficient of constant on both sides. $4A_0 - 13A_1 = 6$.
 $\Rightarrow 4A_0 - 13(2) = 6 \Rightarrow 4A_0 - 26 = 6 \Rightarrow 4A_0 = 32 \Rightarrow A_0 = 8$.
 $\therefore A_0 = 8, A_1 = 2$.
 Substitute $A_0 = 8, A_1 = 2$ in (3), we get
 $a_k^p = (8 + 2k)1^k$.
 $\therefore$ The general solution of given recurrence relation is
 $a_k = a_k^h + a_k^p$.
 $a_k = C_1 2^k + C_2 5^k + (8 + 2k)1^k \rightarrow (4)$.
 Given initial conditions. $a_0 = 1, a_1 = 2$.
 put $n=0$ in (4). $a_0 = C_1 2^0 + C_2 5^0 + (8 + 2(0))1^0$.
 $C_1 + C_2 = -7 \rightarrow (5)$.
 put $n=1$ in (4). $a_1 = C_1 2^1 + C_2 5^1 + (8 + 2(1))1^1$.
 $2C_1 + 5C_2 + 10 = 2$.
 $2C_1 + 5C_2 = -8 \rightarrow (6)$.

$2 \times (5) - (6)$
 $2C_1 + 2C_2 = 8 - 14$
 $2C_1 + 5C_2 = -8$
 $-3C_2 = 16$
 $C_2 = -\frac{16}{3} \Rightarrow C_2 = -\frac{16}{3}$
 Sub $C_2 = -\frac{16}{3}$ in (5). $C_1 + C_2 = -7$
 $\Rightarrow C_1 - \frac{16}{3} = -7 \Rightarrow C_1 = -7 + \frac{16}{3} = -\frac{21}{3} + \frac{16}{3} = -\frac{5}{3}$
 $\therefore C_1 = -\frac{5}{3}$
 $\therefore$ The required general solution of the given recurrence relation is
 $a_k = -\frac{5}{3}(2^k) - \frac{16}{3}(5^k) + (8 + 2k)1^k$

10

(a) Solve the recurrence relation

$$a_n = 4a_{n-1} - 4a_{n-2} + (n+1)2^n$$

A) <https://youtu.be/mwzTx6j0bQg?si=oWuDIOXcjXCzeqQ7>

B) <https://youtu.be/YOIwtgIUdKo?si=45itkulB1-0QiGp6>

Given that $a_n - 4a_{n-1} + 4a_{n-2} = (n+1)2^n \rightarrow (1)$
 To find CF, Consider $a_n - 4a_{n-1} + 4a_{n-2} = 0$ which is a Homogeneous
 linear recurrence relation of degree 2.
 The characteristic equation is $c(m) = 0$.

$$m^2 - 4m + 4 = 0$$

$$(m-2)^2 = 0$$

$$m = 2, 2$$

The characteristic roots are 2, 2 which are real and equal.

$$\text{Complementary function} = a_n^h = (C_1 + C_2 n)2^n \rightarrow (2)$$

To find P.I., R.H.S of (1) is $(n+1)2^n$ where 2 is a characteristic
 root of multiply it by n^2 .

$$\therefore a_n^p = (A_0 + A_1 n)2^n \cdot n^2 \rightarrow (3)$$

$$\text{Consider } a_n - 4a_{n-1} + 4a_{n-2} = (n+1)2^n \quad (1)$$

$$(A_0 + A_1 n)2^n n^2 - 4(A_0 + A_1(n-1))2^{n-1}(n-1)^2 + 4(A_0 + A_1(n-2))2^{n-2}(n-2)^2 = (n+1)2^n$$

$$\Rightarrow A_0 2^n n^2 + A_1 2^n n^3 - 4(A_0 + A_1(n-1))2^{n-1}(n^2 - 2n + 1) + 4(A_0 + A_1(n-2))2^{n-2}(n^2 - 4n + 4) = 2^n(n+1)$$

$$\Rightarrow 2^n [A_0 n^2 + A_1 n^3 - 4(A_0 + A_1(n-1))2^{n-1}(n^2 - 2n + 1) + 4(A_0 + A_1(n-2))2^{n-2}(n^2 - 4n + 4)] = 2^n(n+1)$$

$$\Rightarrow (A_0 n^2 + A_1 n^3) - (2A_0 + 2A_1 n - 2A_1)(n^2 - 2n + 1) + (A_0 + A_1(n-2))(n^2 - 4n + 4) = (n+1)$$

$$\Rightarrow n^3 A_1 + A_0 n^2 - 2A_0 n^2 + 4A_0 n - 2A_0 - 2A_1 n^3 + 4A_1 n^2 - 2A_1 n + 2A_1 n^2 - 4A_1 n + 2A_1 + A_0 n^2 - 4n A_0 + 4A_0 + A_1 n^3 - 4n^2 A_1 + 4A_1 n - 2A_1 n^2 + 8A_1 n - 8A_1 = n+1$$

$$\Rightarrow (4A_0 - 2A_1 - 4A_1 - 4A_0 + 8A_1)n + (-2A_0 + 2A_1 + 4A_0 - 8A_1 + 4A_1) = n+1$$

$$\Rightarrow (4A_1)n + (2A_0 - 8A_1) = n+1$$

$$\text{Compare coefficient of } n \text{ on both sides } 4A_1 = 1 \Rightarrow A_1 = 1/4$$

$$\text{Compare coefficient of constant on both sides } 2A_0 - 8A_1 = 1 \Rightarrow 2A_0 - 8(1/4) = 1 \Rightarrow 2A_0 = 1 + 2 \Rightarrow A_0 = 3/2$$

$\therefore$ Substitute A_0 & A_1 in (3), we get

$$a_n^p = (1 + 1/4 n)2^n \cdot n^2 \rightarrow (4)$$

Substitute (2) & (4) in General solution of (1) that is $a_n = a_n^h + a_n^p$.

The required general solution of the given recurrence relation is

$$a_n = (C_1 + C_2 n)2^n + (1 + 1/4 n)2^n n^2$$

(b) Solve the recurrence relation $F_n = 10F_{n-1} - 25F_{n-2}$.

	(Q) 10b Solve $F_n = 10F_{n-1} + 25F_{n-2}$. Given that $F_n - 10F_{n-1} + 25F_{n-2} = 0$, which is a homogeneous linear recurrence relation of degree 2. The characteristic equation is $C(m) = 0$. $m^2 - 10m + 25 = 0$ $m^2 - 5m - 5m + 25 = 0$ $m(m-5) - 5(m-5) = 0$ $(m-5)(m-5) = 0$ $m = 5, 5$ $\therefore$ The characteristic roots are $m = 5, 5$, which are real and equal. $\therefore$ The general solution of recurrence relation is $a_n = \text{Complementary function} = C_1 5^n + C_2 n 5^n$	
11	If $a_0 = 0$, $a_1 = 1$, $a_2 = 4$ and $a_3 = 37$ satisfy the recurrence relation $a_{n+2} + b a_{n+1} + c a_n = 0$ for $n \geq 0$. Determine the constants b and c and then solve the recurrence relation for a_n.	https://youtu.be/prLzCxn8Yvo?si=bCajHGMBxw9yzUDx

Ex 1 Given Recurrence Relation is

$a_{n+2} + ba_{n+1} + ca_n = 0$ for $n \geq 0$. $\rightarrow$ ①
 given initial conditions are $a_0 = 0, a_1 = 1, a_2 = 4$ and $a_3 = 37$.
 put $n = 0$ in ①, we get

$$a_2 + ba_1 + ca_0 = 0 \rightarrow ②$$

Substitute given values of a_0, a_1, a_2 in ②, we get

$$4 + b(1) + c(0) = 0$$

$$\boxed{b = -4}$$

Sub $n = 1$ in ①, we get

$$a_3 + ba_2 + ca_1 = 0 \rightarrow ③$$

Sub given initial conditions, we get $(37) + (-4)(4) + c(1) = 0$

$$\Rightarrow c = -37 + 16 = -21$$

$$\therefore \boxed{c = -21}$$

$\therefore$ Substitute $b = -4$ & $c = -21$ in eqⁿ ①, recurrence relation of a_n becomes

$a_{n+2} - 4a_{n+1} - 21a_n = 0$ which is homogeneous recurrence relation of degree 2.

The characteristic equation of ④ is $C(m) = 0$.

$$\Rightarrow m^2 - 4m - 21 = 0$$

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{4 \pm \sqrt{4^2 - 4(1)(-21)}}{2}$$

$$= \frac{4 \pm \sqrt{16 + 84}}{2}$$

$$= \frac{4 \pm \sqrt{100}}{2}$$

$$= \frac{4 \pm 10}{2} = \frac{4+10}{2}, \frac{4-10}{2} = \frac{14}{2}, \frac{-6}{2} = 7, -3$$

$\therefore$ The General solution of given recurrence relation is

$$\boxed{a_n = C_1 7^n + C_2 (-3)^n} \rightarrow ⑤$$

Given that $a_0 = 0, a_1 = 1, a_2 = 4, a_3 = 37$.

put $n = 0$ in ⑤, we get

$$a_0 = C_1 + C_2 \Rightarrow C_1 + C_2 = 0 \Rightarrow \boxed{C_1 = -C_2} \quad (a_0 = 0)$$

put $n = 1$ in ⑤, we get

$$a_1 = C_1 7 + C_2 (-3) \Rightarrow -C_2 7 - 3C_2 = 1 \quad (a_1 = 1)$$

$$\Rightarrow -10C_2 = 1 \Rightarrow \boxed{C_2 = -\frac{1}{10}} \quad \text{Sub in } C_1, \text{ we get } \boxed{C_1 = \frac{1}{10}}$$

Substitute $C_1 = \frac{1}{10}$ & $C_2 = -\frac{1}{10}$ in ⑤.

The required general solution of given recurrence relation is

$$a_n = \frac{1}{10} (7^n) - \frac{1}{10} (-3)^n$$

$$\therefore \boxed{a_n = \frac{1}{10} (7^n - (-3)^n)} \quad (\text{you can check Sub } n=3 \text{ in } a_n \text{ we get } a_3 = \frac{1}{10} (7^3 - (-3)^3) = \frac{(343 + 27)}{10} = 37)$$

- 12 (a) Solve the recurrence relation $a_n = 10a_{n-1} + 29a_{n-2}$ for $n \geq 3$.
 Given that $a_0 = 10, a_1 = 100$.

- A) <https://youtu.be/e51BzJYqMqY?si=Yp2vkzMrJMzyJuIn>
 B) <https://youtu.be/vdiY7aPA4dI?si=kU7TEJi1bwcb7M3o>

Q12b) Given that $a_n = 10a_{n-1} + 29a_{n-2}$ for $n \geq 2$,
 $a_n - 10a_{n-1} - 29a_{n-2} = 0$, which is a homogeneous linear recurrence relation of degree 2.

The characteristic equation is $c(m) = 0$.

$m^2 - 10m - 29 = 0$, which is a quadratic equation.

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{10 \pm \sqrt{100 - 4(-29)}}{2(1)}$$

$$= \frac{10 \pm \sqrt{100 + 116}}{2}$$

$$= \frac{10 \pm \sqrt{216}}{2}$$

$$= \frac{10 \pm 6\sqrt{6}}{2} = 5 \pm 3\sqrt{6}, \frac{10 - 6\sqrt{6}}{2}$$

$$= 5 \pm 3\sqrt{6}, 5 - 3\sqrt{6}$$

The characteristic roots are $5 + 3\sqrt{6}$, $5 - 3\sqrt{6}$ real & distinct

The general solution of the given recurrence relation

$$x_n = \text{Complementary function} = C_1(5 + 3\sqrt{6})^n + C_2(5 - 3\sqrt{6})^n \rightarrow (1)$$

Given initial conditions are $a_0 = 10$, $a_1 = 100$.

Substitute $n=0$ in (1),

$$a_0 = C_1 + C_2 \Rightarrow C_1 + C_2 = 10 \rightarrow (2)$$

Substitute $n=1$ in (1)

$$a_1 = C_1(5 + 3\sqrt{6}) + C_2(5 - 3\sqrt{6})$$

$$C_1(5 + 3\sqrt{6}) + C_2(5 - 3\sqrt{6}) = 100$$

$$5(C_1 + C_2) + 3\sqrt{6}(C_1 - C_2) = 100 \quad (\text{from } (2) \ C_1 + C_2 = 10)$$

$$5(10) + 3\sqrt{6}(C_1 - C_2) = 100$$

$$3\sqrt{6}(C_1 - C_2) = 100 - 50$$

$$C_1 - C_2 = \frac{50}{3\sqrt{6}} \rightarrow (3)$$

Adding (2) & (3), we get

$$2C_1 = 10 + \frac{50}{3\sqrt{6}}$$

$$C_1 = \frac{25}{3\sqrt{6}} + 5$$

Substitute C_1 in (2), we get

$$C_2 = 10 - C_1 = 10 - 5 - \frac{25}{3\sqrt{6}} = 5 - \frac{25}{3\sqrt{6}}$$

Substitute C_1 & C_2 values in (1).

$\therefore$ The required general solution of the given recurrence relation

$$\text{is } a_n = \left(5 + \frac{25}{3\sqrt{6}}\right)(5 + 3\sqrt{6})^n + \left(5 - \frac{25}{3\sqrt{6}}\right)(5 - 3\sqrt{6})^n$$

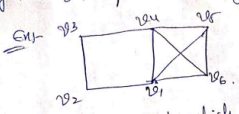
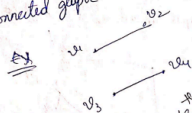
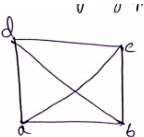
(b) Find the coefficient of x^{10} in $(x^3 + x^4 + \dots)^2$

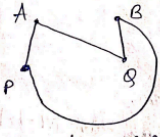
$$\begin{aligned}
 &\text{Given } (x^3 + x^4 + \dots)^\infty \\
 &= x^3 (1 + x + x^2 + \dots)^\infty \\
 &= (x^3) (1-x)^{-1} \\
 &= x^3 (1-x)^{-2} \\
 &= x^3 (1 + 2x + 3x^2 + \dots) \\
 &= x^3 \left(\sum_{n=0}^{\infty} (n+1)x^n \right) \\
 &= \sum_{n=0}^{\infty} (n+1)x^{n+3} \\
 &\text{The coeff of } x^{10} \text{ is } n+1 \\
 &\text{compare } x^{n+3} \text{ with } x^{10} \\
 &n+3=10 \\
 &n=7 \\
 &\therefore \text{the coeff of } x^{10} \text{ is } 7+1=8.
 \end{aligned}$$

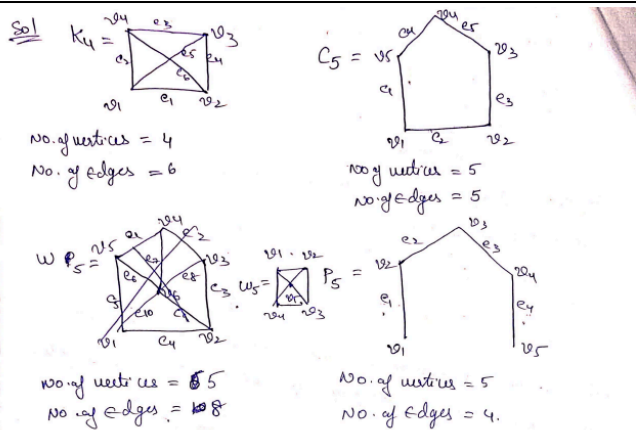
UNIT 5

Graph Theory :

Basic Concepts of Graphs, Matrix Representation of Graphs: Adjacency Matrices, Incidence Matrices, Sub graphs Paths and Circuits, Isomorphic Graphs, Eulerian and Hamiltonian Graphs, Multigraphs, Planar Graphs, Euler's Formula, Graph Colouring and Covering, Chromatic Number, Kruskal's and Prim's Algorithm for finding minimal spanning trees. (Problems Only and Theorems without Proofs)

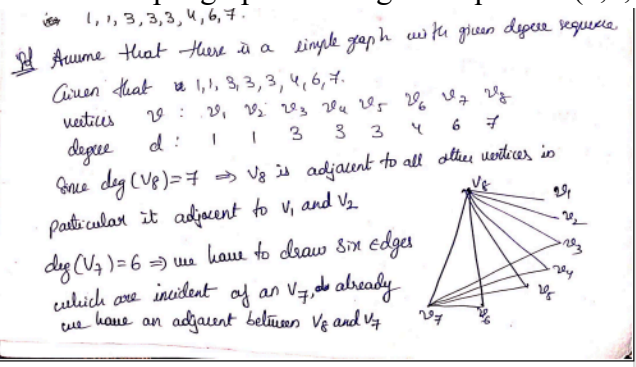
1	What is connected graph give an example for disconnected graph. <u>Connected graph</u>: A non directed graph G is said to be connected graph if there exists a path between any pair of vertices in G.  <u>Disconnected graph</u>: A graph which is not connected is called a disconnected graph.  There is no path between these vertices, so it is disconnected.	https://youtu.be/8wGvvUHMVYk?si=bkpiIgRr8gAyhKR
2	Give an Example of a Graph with Hamilton cycle but no Euler Circuit.  In the above graph Hamiltonian cycle is a-b-c-d-a. But there is no Eulerian circuit (because degree of all vertices are odd).	https://youtu.be/MLalI_OMOdk?si=ANrPcA90wJsV_ahS
3	A simple graph contains 35 edges, four vertices of degree 5, five vertices of degree 4 and four vertices of degree 3. Find the number of vertices with degree 2.	https://youtu.be/-CexpKYUGsQ?si=4NK6eJRsyAWnA1CJ

	<u>Sol</u> Given that no. of edges = $35 = E$ 4 vertices of degree 5 $\Rightarrow 4 \times 5 = 20$ 5 vertices of degree 4 $\Rightarrow 5 \times 4 = 20$ 4 vertices of degree 3 $\Rightarrow 4 \times 3 = 12$ Let the no. of vertices with degree 2 = $x \Rightarrow 2x$ We know that sum of degree of vertex = $2E$ $\Rightarrow 20 + 20 + 12 + 2x = 2(35) = 70$ $\Rightarrow 52 + 2x = 70$ $\Rightarrow 2x = 70 - 52 = 18$ $\Rightarrow \boxed{x = 9}$ There are 9 no. of vertices with degree 2.	
4	Write any two advantages of Adjacency matrix (4/5/24) <u>Sol</u> - write any two advantages of Adjacency Matrix. <u>Sol</u> - Two advantages of adjacency matrix are (1) Easy to represent and understand (2) Quick to check if an edge exists between two vertices.	https://youtu.be/y6EgXkswWZQ?si=LYM3gsvtLx9AL8zj
5	Show that $K_{m,n}$ is planar if either $m \leq 2$ or $n \leq 2$. <u>Sol</u> Given that $K_{m,n}$ if $m \leq 2$ or $n \leq 2$ means we have 3 possibilities (1) $m=1, n=1$ (2) $m=2, n=1$ (3) $m=2, n=2$ (1) $K_{m,n}$ if either $m=1$ or $n=2$ $K_{1,2} =$  There is no edge crossing clearly $K_{1,2}$ is planar (2) $K_{m,n}$ if $m=2, n=1$ $K_{2,1} =$  There is no edge cross, $K_{2,1}$ is planar graph (3) $K_{m,n}$ if $m=2, n=2$ $K_{2,2} =$  It is a non-planar. but we convert it into planar by changing the edge direction. $\times$ $K_{2,2} =$  It is planar graph (since no there is no edge cross). $\therefore K_{m,n}$ is a planar graph if either $m \leq 2$ or $n \leq 2$.	https://youtu.be/9PdPlk9ahjY?si=BCEYQ7ExrGSRp1L1
6	Find the Number of vertices and edges in K_4, C_5, P_5, W_5.	https://youtu.be/-VsTYT8QsmY?si=spNY9oqMQRpyCxp7



7 Is there a simple graph with degree sequence (1,1,3,3,3,4,6,7)?

https://youtu.be/H1h0d_SSsq4?si=Tva_5G3V3mDcXIgW



to we need to draw remaining 5 edges

we cannot draw loop at v_7 because we assume that graph is simple

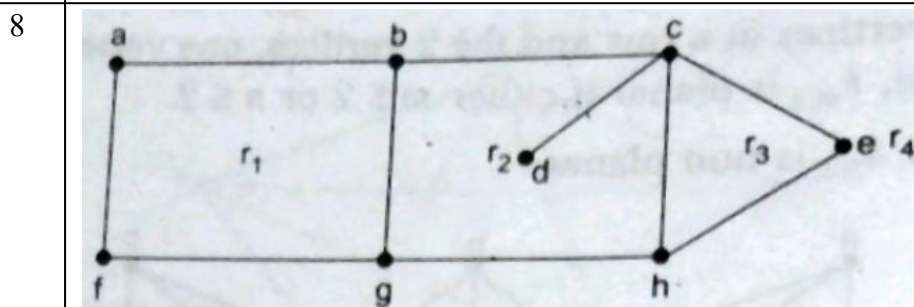
we can not draw edges from v_7 to v_1 & v_2 because $\deg(v_1) = 1$ and $\deg(v_2) = 1$.

we have $\deg(v_7) = 5$ which is a contradiction.

Since $\deg(v_7) = 6$.

So our assumption is wrong.

There is no simple graph with the degree sequence 1, 1, 3, 3, 3, 4, 6, 7.



<https://youtu.be/9PdPlk9ahjY?si=UC4FCP--pngs66J2>

Find degree of each region and Verify Euler's Formula for the above Plane Graph

Sol: Given graph is a connected planar graph which contains $|V| = 8$, $|E| = 10$.
 Number of Regions $|R| = |E| - |V| + 2 = 10 - 8 + 2 = 4$.
 Here, Region R_1 is bounded by the cycle $(a-b-g-f-a) \Rightarrow \deg(R_1) = 4$.
 Here, Region R_2 is bounded by the cycle $(b-c-d-e-h-g-b) \Rightarrow \deg(R_2) = 6$.
 Region R_3 is bounded by the cycle $(c-e-h-c) \Rightarrow \deg(R_3) = 3$.
 outer Region R_4 is bounded by the cycle $(a-b-c-e-h-g-f-a) \Rightarrow \deg(R_4) = 7$.
 Verification of Euler's formula:

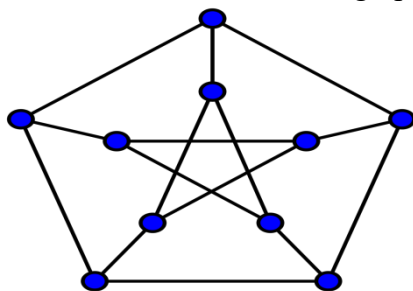
$$|V| - |E| + |R| = 2$$

$$\Rightarrow 8 - 10 + 4 = 2$$

$$\Rightarrow 2 = 2$$

$$\therefore \text{Euler's formula is verified for the given graph.}$$

9 (a) Find the chromatic number of the Peterson graph:



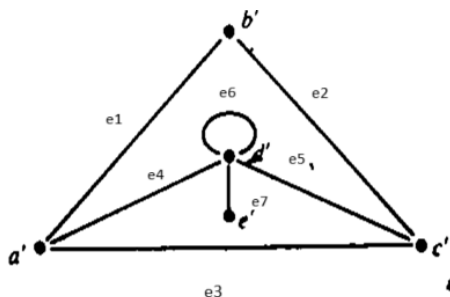
Sol:- we know that peterson graph contains 10 vertices & 15 edges.

Let $C_1 = \text{Red}$
 $C_2 = \text{Green}$
 $C_3 = \text{Blue}$

vertices	a	b	c	d	e	f	g	h	i	j
degree	3	3	3	3	3	3	3	3	3	3
colour	C_1	C_3	C_2	C_1	C_2	C_3	C_2	C_1	C_3	C_2

The minimum number of colours required to colour all the vertices of graph G is 3.
 $\therefore$ the chromatic number of given peterson graph is $\chi(G) = 3$.

(b) Define adjacency and incidence matrix and write adjacency and incidence matrix of the following graph



A) <https://youtu.be/2KFVnXwbKcs?si=x8sTMXMyDJDeZfyf>
 B) https://youtu.be/y6EgXkswWZQ?si=7Us4l_oJxdfqE2qe

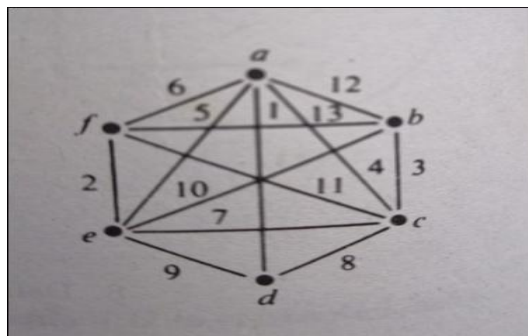
Q1) The adjacency matrix for given undirected graph is

$$A_G = \begin{matrix} & \begin{matrix} a' & b' & c' & d' & e' \end{matrix} \\ \begin{matrix} a' \\ b' \\ c' \\ d' \\ e' \end{matrix} & \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

The incidence matrix for the given undirected graph is

$$I_G = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 \end{matrix} \\ \begin{matrix} a' \\ b' \\ c' \\ d' \\ e' \end{matrix} & \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

10 Apply prim's algorithm for the below graphs to find minimal spanning tree.



Ans:- Weighted matrix for given graph

$$\begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{bmatrix} \infty & 12 & 13 & 1 & 5 & 6 \\ 12 & \infty & 3 & 4 & 10 & 2 \\ 13 & 3 & \infty & 8 & 7 & 11 \\ 1 & 4 & 8 & \infty & 9 & 1 \\ 5 & 10 & 7 & 9 & \infty & 2 \\ 6 & 2 & 11 & 1 & 2 & \infty \end{bmatrix} \end{matrix}$$

Step:- Since the graph contains 6 vertices, so we need to select 5 edges to construct minimal spanning tree.

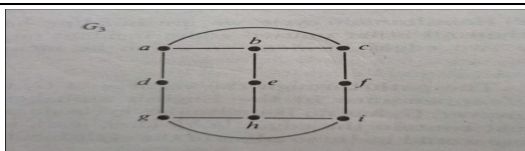
S.No.	Eligible edge with weights	Selected edge with weights
1.	ab(12), ac(13), ad(1), ae(5), af(6)	ad(1)
2.	dc(8), de(9), db(4), ae(5), af(6)	ac(4)
3.	cb(3), cf(2), ce(7), cd(8)	cb(3)
4.	be(10), bf(2), bc(10)	ae(5)
5.	cf(2)	ef(2)

Minimal spanning tree for given graph

Total weight of minimal spanning tree = 1 + 2 + 3 + 4 + 5 = 15.

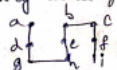
A) https://youtu.be/dtvoe_Ixu_m4?si=01ELqeh7208_pwND
 B) <https://youtu.be/mNYvM782WUc?si=evrdQHh4SXG9dwni>

b) Determine whether the following graph is Hamiltonian ?

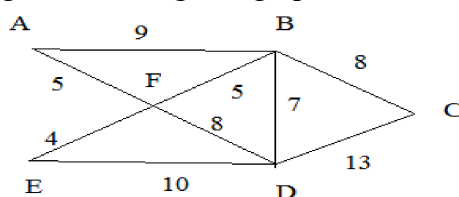


Sol Since degree of each vertices of d, e, f are 2, so we need to include all the 6 edges a-d-g, b-e-h, c-f-i in hamiltonian cycle. Now we consider a vertex b. Here we include edge a-b or b-c in hamiltonian cycle. Since b-e already included. let us include a-b then we need to delete unnecessary edge b-c. now the degree of c is 2 we need to include edge c-a in hamiltonian cycle but then would be degree of a is 3. So there is no hamiltonian cycle in graph. $\therefore$ The given graph is not hamiltonian.

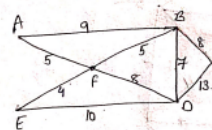
hamiltonian path of above graph is



11 Write Kruskal's algorithm to find minimal spanning tree and find minimal spanning tree for the given graph.



Sol (write above 3 steps. then write following)



Step	Edge	Cost	Edge included in T (Yes/No)
1.	E-F	4	Yes
2.	F-B	5	Yes
3.	A-F	5	Yes
4.	B-D	7	Yes
5.	B-C	8	Yes
6.	F-D	8	No
7.	A-B	9	No
8.	E-D	10	No
9.	D-C	13	No

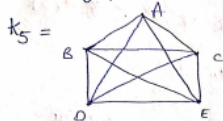


Step 2: Minimum spanning tree, since graph contain 6 vertices its minimum spanning tree must contain 5 edges.

Cost = Total weight of Minimum Spanning tree = $4 + 5 + 5 + 7 + 8 = 29$.

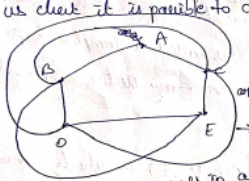
b) Prove that K_5 is not planar

Sol Complete graph with 5 vertices.



(Complete graph mean each vertex is adjacent to all the other vertices in G)

let us check it is possible to draw planar graph or not

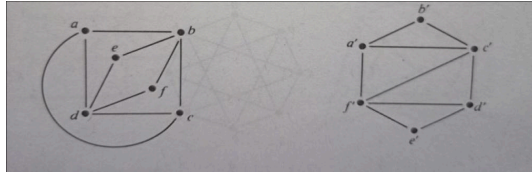


Since edges are cross in all possible way. $\therefore$ On both ways it is not possible to draw plane edge which not cross with other. $\therefore$ The graph K_5 is not planar graph.

A) <https://youtu.be/ToACXUBwouc?si=UyqdgY-E89VSjmPe>

B) <https://youtu.be/DlGQ9F6iVb8?si=H1vszUZ4kE1m3mw8>

a) Verify G and G' are isomorphic or not?



Sol

Step 1:- No. of vertices in $G = 6$
 No. of vertices in $G' = 6$
 $\therefore$ The number of vertices in both graphs G and G' are same.

Step 2:- No. of ~~total~~ edges in $G = 9$
 No. of edges in $G' = 9$
 $\therefore$ The number of edges in both graphs G and G' are same (unlike more to the next step otherwise stop here conclude they are not isomorphic)

Step 3:- Now we need to calculate degree sequence for each graphs.

In G	In G'
$\deg(a) = 3$	$\deg(a') = 3$
$\deg(b) = 4$	$\deg(b') = 2$
$\deg(c) = 3$	$\deg(c') = 4$
$\deg(d) = 4$	$\deg(d') = 3$
$\deg(e) = 2$	$\deg(e') = 2$
$\deg(f) = 2$	$\deg(f') = 4$

degree sequence of graph G in ascending order $= (2, 2, 3, 3, 4, 4)$
 degree sequence of graph G' in ascending order $= (2, 2, 3, 3, 4, 4)$
 $\therefore$ Degree sequence is same in both the graphs.

Step 4:- (Observe adjacent vertices of each vertex in G and G')
 one to one correspondence between the vertices of G & G'
 (for each vertex in graph G we have to find an equivalent vertex in G')

From step 3 observe adjacent vertices

$\deg(a) = 3$ and $\deg(a') = 3 \Rightarrow a = a'$ [Just say that there is one to one correspondence between these vertices]

From graph G : $\deg(a) = 3$ is connected to b, c, d
 From graph G' : $\deg(a') = 3$ is connected to b', c', d'

$\deg(b) = 4$ is connected to a, c, d, e
 $\deg(b') = 2$ is connected to a', c'

$\deg(c) = 3$ is connected to a, b, d
 $\deg(c') = 4$ is connected to a', b', d', e'

$\deg(d) = 4$ is connected to a, b, c, f
 $\deg(d') = 3$ is connected to a', b', c'

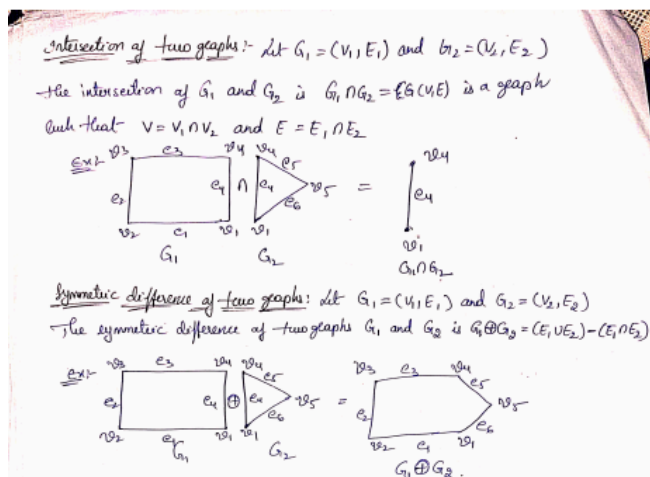
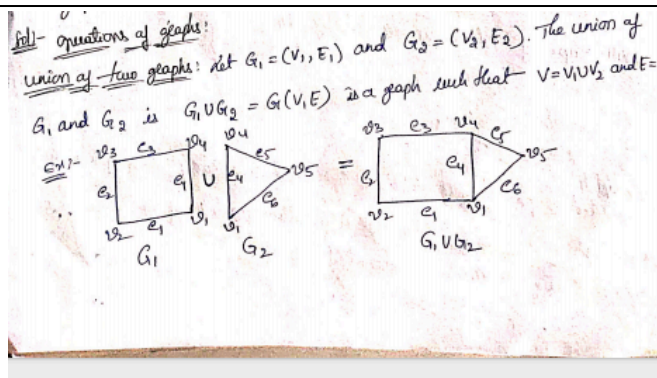
$\deg(e) = 2$ is connected to d, f
 $\deg(e') = 2$ is connected to d', f'

$\deg(f) = 2$ is connected to d, e
 $\deg(f') = 4$ is connected to d', e', c', b'

$\therefore$ There is no one-to-one correspondence between vertices in both graphs
 $\therefore$ The given graphs G and G' are not isomorphic.

b) Explain union intersection and symmetric difference between the graphs with suitable Examples.

A) <https://youtu.be/44ZPS0r5INw?si=OjBiJe8RrbU771f1>
 B) <https://youtu.be/PxffSUQRkG4?si=Hd5crTDT7Gnap0dR>



Text Books:

1	J.P. Tremblay and R. Manohar, Discrete Mathematical Structures with Applications to Computer Science, Tata McGraw Hill, 2002.
2	1. Joe L. Mott, Abraham Kandel and Theodore P. Baker, Discrete Mathematics for Computer Scientists & Mathematicians, 2nd Edition, Pearson Education.

Reference Books:

1	C. L. Liu and, Elements of Discrete Mathematics-A Computer Oriented Approach
2	Narsingh Deo, Graph Theory with Applications to Engineering and Computer Science.
3	Dr. D S Chandrasekharaiah, Mathematical Foundations of Computer Science, Prism Book Pvt Ltd.
4	S. K. Chakraborty and B.K. Sarkar, Discrete Mathematics Oxford 2011.